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United States Patent	12408719
Kind Code	B2
Date of Patent	September 09, 2025
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Protective apparel system with a lens assembly

Abstract

A protective apparel system comprising a surgical helmet including a peripheral device and a surgical garment configured to be disposed over the helmet. The garment comprises a fabric and a face shield defining a sterile barrier between the wearer and the environment. At least one of a coupling member or housing may disposed on the face shield, the coupling member configured for removably couple the peripheral device to the face shield to orient the peripheral device relative to the face shield when the surgical garment is disposed over the surgical helmet. The housing may removably couple a lens assembly to the face shield.

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Appl. No.:	18/627796
Filed:	April 05, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240245159 A1	Jul. 25, 2024

Related U.S. Application Data

continuation parent-doc US 17255630 US 11969046 WO PCT/US2019/039523 20190627 child-doc US 18627796
us-provisional-application US 62690559 20180627

Publication Classification

Int. Cl.: A42B3/04 (20060101); A41D13/11 (20060101); A41D13/12 (20060101)

U.S. Cl.:

CPC A42B3/044 (20130101); A41D13/1161 (20130101); A41D13/1218 (20130101);

Field of Classification Search

CPC: A42B (3/044); A42B (1/242); A42B (1/244); A41D (13/1184); A41D (13/1218); A62B (17/04); A61F (9/06); A61B (1/0692); G02B (27/0176); G02B (27/024)

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/255,630, filed on Dec. 23, 2020, which is a national stage of PCT International Patent Application No. PCT/US2019/039523, filed on Jun. 27, 2019, which claims priority to and the benefit of U.S. Provisional Application No. 62/690,559, filed on Jun. 27, 2018, which are hereby incorporated herein by reference in their entirety.

BACKGROUND

(1) Protective apparel systems are used in surgical procedures to provide a sterile barrier between the medical/surgical personnel and the patient. The protective apparel system includes a helmet that supports a surgical garment. The helmet may include a headlamp to light a portion of a patient or workspace as the medical/surgical personnel operates on the patient. This assemblage is worn by medical/surgical personnel to establish the sterile barrier.

(2) The sterile barrier prevents the exchange and/or transfer of particles or foreign material during the surgical procedure or examination. In addition, the barrier serves to protect the wearer from exposure to blood and other bodily fluids. Maintaining a reliable barrier between the wearer and the patient is of the utmost importance.

(3) During medical and surgical procedures, a wearer may need to adjust or redirect the light from the headlamp for better visibility and lighting of the workspace. Adjustments or redirections of the light from the headlamp require adjustment, tilt, or manipulation of the headlamp. For example, the headlamp may be tilted downwards to illuminate the patient's body. Currently, the wearer adjusts the headlamp through the surgical garment. At the same time, maintaining visibility of the surgical workspace is paramount. By tilting the headlamp downwards, the visibility of the workspace is decreased. Thus, any adjustment of the headlamp during the procedure should not compromise visibility of the procedure.

(4) Furthermore, there may be instances where the headlamp position, while preoperatively acceptable, becomes unacceptable during the procedure. In such instances, the procedure may be delayed until proper adjustment of the headlamp is obtained. This decreases the efficiency of the procedure by delaying the procedure.

(5) A protective apparel system with features designed to overcome at least the aforementioned challenges is desired. These and other configurations, features, and advantages of the present disclosure will be apparent to those skilled in the art. The present disclosure is not to be limited to or by these configurations, features, and advantages.

BRIEF SUMMARY

(6) One configuration of a wearable surgical garment is provided for use with a surgical helmet having a headlamp. The surgical garment comprises a surgical fabric configured to provide a microbial barrier between a medical environment and a wearer. The surgical garment defines an environment side and a wearer side. A coupling member is provided on the wearer side. A lens assembly is coupled to the surgical fabric on the environment side. The coupling member may function to couple to the headlamp on the wearer side such that the headlamp is aligned to the lens assembly. The lens assembly aligns with the headlamp such that the light emitted by the headlamp is transmitted through the lens assembly. The lens assembly comprises a first lens and a control member wherein the control member is operatively coupled to the first lens. The control member is manipulatable by the wearer on the environment side of the surgical garment to adjust a characteristic of the lens assembly.

(7) One configuration of a protective apparel apparatus is provided for use with a surgical helmet having a headlamp. The protective apparel apparatus comprises a wearable surgical garment and a lens assembly. The wearable surgical garment defines an environment side and a wearer side. The

surgical garment comprises a surgical fabric configured to provide a microbial barrier between a medical environment and a wearer. The lens assembly is coupled to the surgical garment on the environment side and configured to be aligned with the headlamp. The lens assembly comprises a housing, a first lens coupled to the housing, and a control member coupled to the housing. The control member is operatively coupled to the first lens. The control member is manipulatable by the wearer on the environment side of the surgical garment to adjust a characteristic of the lens assembly.

(8) One configuration of a protective apparel system is provided. A protective apparel system comprises a surgical helmet to be worn over the head of a wearer. The surgical helmet comprises a headlamp. The protective apparel system comprises a surgical garment configured to be at least partially disposed over the surgical helmet. The surgical garment defines an environment side and a wearer side. The surgical garment comprises a surgical fabric configured to provide a microbial barrier between a medical environment and a wearer. A coupling member is provided on the wearer side. The surgical garment further comprises a lens assembly coupled to the surgical garment on the environment side. The lens assembly is configured to be aligned with the headlamp such that the light emitted by the headlamp is transmitted through the lens assembly. The lens assembly comprises a housing, a first lens coupled to the housing, a control member coupled to the housing, and a second lens. The control member is operatively coupled to the second lens and manipulatable by the wearer on the environment side of the surgical garment to adjust a characteristic of the lens assembly. The manipulation of the control member by the wearer on the environment side of the surgical garment adjusts the position of the second lens relative to the first lens to direct the light emitted from the headlamp without breaching the sterile barrier.

(9) The surgical garment, protective apparel apparatus, and the protective apparel system provide solutions to problems that have not been addressed by prior techniques. The techniques described herein provide a solution for adjusting the light from a headlamp without breaching the sterile barrier, which maintains the sterility of the procedure. Thus, the surgical garment, protective apparel apparatus and the protective apparel system provide a solution to optimally satisfy this delicate balance involved with headlamp adjustments and maintaining a sterile barrier. Moreover, any detrimental consequences on the procedure resulting are mitigated because the techniques ensure proper adjustment of the light from the headlamp as well as maintaining the sterile barrier during a procedure. Accordingly, the techniques herein provide a solution to adjustment of light from a headlamp thereby increasing probability of a successful surgical procedure and eliminating breaching of the sterile barrier during the surgical procedure.

(10) The surgical garment, protective apparel apparatus and the protective apparel system may provide solutions other than those described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Referring now to the drawings, exemplary illustrations are shown in detail. Although the drawings represent schematic configurations, the drawings are not necessarily to scale and certain features may be exaggerated to better illustrate and explain an innovative aspect of an illustrative configuration. Further, the exemplary illustrations described herein are not intended to be exhaustive or otherwise limiting or restricting to the precise form and configuration shown in the drawings and disclosed in the following detailed description.

(2) Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

(3) FIG. 1A is a perspective view of an exemplary configuration of a protective apparel system that

includes a surgical helmet, a surgical garment, and a lens assembly, with the surgical helmet shown in phantom.

(4) FIG. 1B is a perspective view of a first configuration of the surgical helmet of the protective apparel system of FIG. 1A, the surgical helmet including a headlamp.

(5) FIG. 1C is a perspective view of a second configuration of the surgical helmet of the protective apparel system of FIG. 1A, the surgical helmet including a headlamp.

(6) FIG. 2A is a perspective view of a first configuration of a surgical garment of the protective apparel system of FIG. 1A.

(7) FIG. 2B is a perspective view of a second configuration of a surgical garment of the protective apparel system of FIG. 1A, the face shield of the surgical garment including an opening.

(8) FIG. 2C is a perspective view of a third configuration of a surgical garment of the protective apparel system of FIG. 1A, the surgical garment including a lens assembly comprising a housing and a control member.

(9) FIG. 3A is a partially exploded view of an exemplary configuration of the protective apparel system of FIG. 1A including a surgical helmet with a headlamp and a first configuration of a lens assembly for use with the face shield of the first configuration of the surgical garment of FIG. 2A.

(10) FIG. 3B is a partially exploded view of an exemplary configuration of the protective apparel system of FIG. 1A including a surgical helmet with a headlamp and a second configuration of a lens assembly for use with the face shield of the second configuration of the surgical garment of FIG. 2B.

(11) FIG. 3C is a partially exploded view of an exemplary configuration of the protective apparel system of FIG. 1A including a surgical helmet with a headlamp and a third configuration of a lens assembly.

(12) FIG. 4 is a cross-sectional view of a portion of the protective apparel system of FIG. 1A along line A-A.

(13) FIG. 5A is a partially exploded view of an exemplary configuration of a headlamp and a coupling member for use with the first configuration of the surgical garment and the lens assembly of FIG. 3A.

(14) FIG. 5B is a partially exploded view of the first configuration of the lens assembly of FIG. 3A, the lens assembly including a housing, a first lens, a second lens, and a control member.

(15) FIG. 6A is a partially exploded view of an exemplary configuration of a headlamp and a coupling member for use with the second configuration of the surgical garment and the lens assembly of FIG. 3B.

(16) FIG. 6B is a partially exploded view of the second configuration of the lens assembly of FIG. 3B, the lens assembly including a housing having a coupling feature, a first lens, a second lens, and a control member.

(17) FIG. 7A is a cross-sectional view of the layers of a first lens for use with a lens assembly.

(18) FIG. 7B is a cross-sectional view of the layers of a second lens for use with a lens assembly.

(19) FIG. 8A is a perspective view of the second configuration of the lens assembly of FIGS. 3B and 4, illustrating the control member positioned at a first distance away from the housing.

(20) FIG. 8B is a perspective view of the second configuration of the lens assembly of FIGS. 3B and 4, illustrating the control member positioned at a second distance away from the housing.

(21) FIG. 8C is a perspective view of the second configuration of the lens assembly of FIGS. 3B and 4, illustrating the control member positioned at a third distance away from the housing.

(22) FIG. 9A is a schematic view of FIG. 8A illustrating an exemplary light pattern produced by the second configuration of the lens assembly including a first lens and a second lens when the control member is positioned at the first distance away from the housing.

(23) FIG. 9B is a schematic view of FIG. 8B illustrating an exemplary light pattern produced by the second configuration of the lens assembly including a first lens and a second lens when the control member is positioned at the second distance away from the housing.

(24) FIG. 9C is a schematic view of FIG. 8C illustrating an exemplary light pattern produced by the second configuration of the lens assembly including a first lens and a second lens when the control member is positioned at the third distance away from the housing.

(25) FIG. 10 is a schematic view of an exemplary light pattern produced by the third configuration of the light assembly of FIG. 3C.

DETAILED DESCRIPTION

I. Overview of Patient Apparel System

(26) Maintaining a sterile barrier between a medical personnel/healthcare provider and a patient to prevent the exchange and/or transfer of particles or foreign material during a medical procedure or examination is of the utmost importance. The healthcare provider may wear an assembly known as a protective apparel system 5 to maintain the sterile barrier.

(27) Referring to FIG. 1A, an exemplary configuration of a protective apparel system 5, including a protective apparel apparatus 10 and a surgical helmet 22, is illustrated. Accordingly, the protective apparel apparatus 10 may comprise a wearable surgical garment 20 configured to be at least partially disposed over the surgical helmet 22, as shown in FIG. 1A. The surgical garment 20 may include a surgical fabric 24 configured to cover the surgical helmet 22 and at least a portion of the head of the wearer. The protective apparel apparatus 10 may also comprise a shield 26 coupled to the surgical garment 20. The protective apparel apparatus 10 may also comprise a lens assembly 14 that may be coupled to the surgical garment 20 or the shield 26. Collectively, the features of the protective apparel apparatus 10 may be configured to define a sterile barrier between the wearer side and the environment side of the protective apparel apparatus 10. Additional features and variations of the protective apparel apparatus 10 that may form a part of the protective apparel system 5 will be described in greater detail below.

(28) The protective apparel system 5 may include the surgical helmet 22. Referring to FIGS. 1B and 1C, exemplary configurations of the surgical helmet 22 that may be used with the protective apparel apparatus 10 as part of the protective apparel system 5 are illustrated. Each configuration of the surgical helmet 22 may generally include a shell 32, a chin bar 31, and a headband 30. The shell 32 is supported by and located above the headband 30. The shell 32 may be configured in an arcuate shape to fit over the head of the wearer. Other helmet designs are contemplated. Many portions of the shell 32 may be formed to define voids, or open interior spaces. For example, the shell 32 may comprise a center void 34. The center void 34 may be located towards the rear of the shell 32. There may be an intake opening or aperture in the top portion of the shell 32 to provide access to the center void 34. The shell 32 may also include additional voids, such as a front void proximate to the front of the shell 32 and a rear void proximate to the rear of the shell 32. The additional voids may be configured to form duct-like structures or passageways within the shell 32.

(29) The surgical helmet 22 may also include one or more peripheral devices, including, but not limited to, a ventilation assembly, a headlamp 12, a camera, a microphone or other communication device, a cooling device, or combinations thereof. For example, a ventilation assembly may be positioned within the shell 32 of the surgical helmet 22 above the head of a wearer. The ventilation assembly may be configured to draw air into the center void 34 of the shell 32 and disperse it through the duct-like structures or passageways within the shell 32. It is contemplated that the peripheral device may also include a microphone disposed in the chin bar 31. In yet another example, the peripheral device may also include a headlamp 12. The headlamp 12 may be coupled to the shell 32 of the surgical helmet 22. While example configurations of the surgical helmet 22 including a peripheral device are described above, the examples are not intended to be limiting. It is contemplated that the peripheral devices described above may be mounted to and/or attached at various locations and orientations relative to the surgical helmet 22.

(30) Additional features of the protective apparel apparatus 10, the surgical helmet 22 and the peripheral devices may be described in one or more of the following U.S. patents, which are hereby incorporated by reference: U.S. Pat. Nos. 6,481,019; 7,752,682; 9,706,808; 6,973,677; 7,735,156;

7,752,682; 8,234,722; 8,282,234; 8,407,818; 8,819,869; and 9,173,437.

(31) Referring to FIGS. 1B and 1C, exemplary configurations of the surgical helmet 22 for use with a peripheral device, such as a headlamp 12, are illustrated. The headlamp 12 may couple to the surgical helmet 22 in any suitable manner. For example, in one configuration, the headlamp 12 may be integrally formed with the helmet 22. In another configuration, the headlamp 12 may be removably coupled to the helmet 22. In yet another configuration, the headlamp 12 may couple to the helmet 22 via a fastener. The fastener may include a bolt, screw, pin, hinge, bracket, or the like to couple the headlamp 12 to the helmet 22.

(32) The headlamp 12 may include a headlamp body 37. The headlamp body 37 may be formed in any suitable configuration. The headlamp 12 may also comprise a transparent cover 36 and a light source 38 (not shown in FIG. 1B) at least partially disposed within the headlamp body 37. The transparent cover 36 may comprise a lens, screen, shield, or other similar cover mechanism configured to allow light to pass through. The transparent cover 36 may be configured to cover a headlamp opening defined by the headlamp body 37 in order to prevent dust, dirt, or moisture from entering the inside of the headlamp 12, while also allowing light transmitted by the light source 38 to pass through. The light source 38 may comprise one or more light emitting diodes (LEDs). The number and type of light emitting diodes may be determined by the desired beam intensity, beam width, electrical power requirement, heat generation and space availability. Other light sources 38 are also contemplated, such as incandescent filaments and the like. Moreover, depending on the required luminous efficacy and required light colors, the light emitting diodes emitting isochromatic or heterochromatic light may be used as the light source 38. In other configurations, the light source 38 may be from any method of generation of light, such as incandescent, luminescent, combustion, fiber optic, and the like.

(33) In one configuration, the headlamp 12 and/or the light source 38 may be battery powered with an internal battery. In another configuration, the headlamp 12 and/or the light source 38 may be coupled to a controller or an external power supply by leads. For example, the headlamp 12 and/or the light source 38 may be self-powered with an internal power supply or may receive power through a host object, if available.

(34) Referring to FIG. 1B, a first exemplary configuration of the surgical helmet 22A is illustrated. The surgical helmet 22A may include the headlamp 12 coupled directly to the shell 32 of the helmet 22A to assist with illuminating or lighting the surgical workspace. The headlamp 12 may be coupled to the shell 32 of the surgical helmet 22A by a hinge, pin, screw, pivot, bracket, hook and loop, or the like. For example, the headlamp 12 may include a bracket configured to engage a corresponding bracket on the shell 32 of the surgical helmet 22. The corresponding brackets of the shell 32 and the headlamp 12 may be coupled by a screw, pin, or the like, and configured to allow the headlamp 12 to be pivoted relative to the shell 32. Alternatively, the headlamp 12 may comprise a bracket including a ball and the shell 32 may comprise a corresponding cavity or recess configured to form a press fit with the ball of the headlamp 12. It is further contemplated that the headlamp 12 may comprise a bracket that is coupled directly to the shell 32 of the surgical helmet 22 by a screw, bolt, or similar fastener. In this configuration of the headlamp 12, the bracket may comprise a pivot, hinge, or similar mechanism to allow the headlamp 12 to be manipulated relative to the surgical helmet 22 when coupled to the shell 32. While FIG. 1B generally illustrates the headlamp 12 being coupled to the underside of the shell 32 on the front of the surgical helmet 22A, it is contemplated that the headlamp 12 may be coupled at other locations on the surgical helmet 22A. For example, it is also contemplated that the headlamp 12 may be coupled to the headband 30 of the surgical helmet 22A in a similar manner.

(35) Alternatively, it is also contemplated that the headlamp 12 may be removably coupled to the surgical helmet 22B by a separate mounting feature 45. FIG. 1C illustrates a second exemplary configuration of the surgical helmet 22B including a mounting feature 45, such as a bracket, for coupling a peripheral device, such as the headlamp 12, to the surgical helmet 22B. For example, as

illustrated in FIG. 1C, the mounting feature **45** may be coupled to the shell **32** of the surgical helmet **22**. The mounting feature **45** may be coupled to the shell **32** by a screw, bolt, pin, adhesive, hook and loop, or other similar fastener. For example, the mounting feature **45** may be adhered to the shell **32** using an epoxy or similar adhesive. Alternatively, the mounting feature **45** may be coupled to the shell **32** using one or more screws.

(36) The mounting feature **45** may also comprise an attachment portion **47** configured to receive the peripheral device, such as the headlamp **12**. For example, as illustrated in FIG. 1C, the bracket includes a ring-shaped attachment portion **47** configured to receive and/or couple to the headlamp **12**. While the exemplary configuration of the attachment portion **47** is illustrated as being ring-shaped in FIG. 1C, other shapes are contemplated. For example, the attachment portion **47** may be configured in a U-shape, square, or other similar shape. The attachment portion **47** may be sized and/or shaped to create a friction fit with the peripheral device, such as the headlamp **12**.

Alternatively, it is contemplated that the attachment portion **47** may comprise a coupling feature that engages a corresponding feature of the headlamp **12**. For example, the attachment portion **47** may comprise a threaded portion, such as on the interior of the aperture defined by the ring-shape, that is configured to engage a corresponding threaded portion on the exterior of the headlamp **12**. Alternatively, the attachment portion **47** may comprise one or more grooves cut in the interior of the aperture defined by the ring-shape that are configured to engage corresponding tabs and/or protrusions on the exterior of the headlamp **12** to couple the headlamp **12** to the attachment portion **47**. In yet another configuration, it is contemplated that the attachment portion **47** may comprise one or more apertures for receiving a fastener such as a pin bolt, or screw, such that the headlamp **12** may be fastened to the attachment portion **47**.

(37) It is also contemplated that the attachment portion **47** of the mounting feature **45** may be configured to couple to and/or engage the protective apparel apparatus **10**. For example, the attachment portion **47** may be configured to removably engage the face shield **26** of the protective apparel apparatus **10** to orient the protective apparel apparatus **10** and/or the face shield **26** relative to the surgical helmet **22B**. This may include aligning the protective apparel apparatus **10** and/or the face shield **26** with the peripheral device, such as the headlamp **12**, that is attached to the mounting feature **45**.

(38) The mounting feature **45** may further comprise a hinge, pivot, or similar mechanism configured to allow the attachment portion **47** of the mounting feature **45** to be manipulated relative to the shell **32** and or surgical helmet **22B**. For example, the mounting feature **45** may comprise a hinge that allows the attachment portion **47** to be folded/swung out of the way when not needed. This would allow the attachment portion **47** to be moved out of the wearer's field of view when not needed, removing a potential obstruction or distraction from the wearer's field of view during a surgical procedure.

(39) As described above, the protective apparel system **5** may comprise a protective apparel apparatus **10** configured to be at least partially disposed over the surgical helmet **22A**, **22B** when worn by a medical professional. During the procedure, it may be necessary to the headlamp **12** that is coupled to the surgical helmet **22A**, **22B** and is on the wearer side of the barrier defined by the protective apparel apparatus **10**. However, it would be disadvantageous if the medical professional were required to adjust the headlamp **12** during the medical procedure without compromising the sterile barrier. Therefore, a protective apparel apparatus **10** including a lens assembly **14** that is at least partially disposed on the environment side of the sterile barrier defined by the protective apparel apparatus **10** and allows for adjustment of the intensity and/or direction of the light emitted from the headlamp **12** without breaking the sterile barrier defined by the protective apparatus **10** is disclosed.

(40) Referring to FIGS. 2A-2C, various exemplary configurations of the protective apparel apparatus **10A**, **10B**, **10C** for use with the protective apparel system **5** of FIG. 1A are illustrated. It will be understood that the protective apparel apparatus **10A**, **10B**, **10C** may be configured to

define an environment side and a wearer side. The barrier may substantially eliminate the likelihood that the wearer may be exposed to the fluid or solid particles of matter from the patient that may be generated during the course of a surgical procedure. The barrier may substantially prevent the transfer of any foreign particles emitted by the wearer from being transferred to the patient during the surgical procedure. The protective apparel apparatus **10** may comprise a surgical garment **20** comprises a surgical fabric **24** configured to at least partially define a barrier, such as a microbial barrier, between the wearer and the surrounding environment. The surgical garment **20** may be configured as a hood or a toga to be placed over the surgical helmet **22A**, **22B**. In one configuration, as shown in FIGS. **1** and **2A**, the surgical garment **20** may be a hood. A hood refers to a surgical garment **20** that covers the head and likely only extends a short distance below the neck when worn by the wearer. Moreover, the surgical garment **20** may be positioned over the surgical helmet **22** and configured to encompass the surgical helmet **22A**, **22B** and, correspondingly, the head of the person wearing the system **5**, thereby covering the wearer's face and back of the head. Alternatively, the surgical garment **20** may be a toga, a shirt, or a jacket. In another instance, the toga refers to the surgical garment **20** that covers the head in the same manner as a hood and extends to at least the waist when worn by the wearer.

(41) The surgical garment **20** may be manufactured from any suitable surgical fabric **24** or combinations of fabrics to help repel and/or absorb water, debris, and other contaminants. The surgical fabric **24** may include multiple layers. One such layer may be a microporous film that allows gas to pass through the fabric **24** while maintaining the microbial barrier. It is further contemplated that the surgical garment **20** may be constructed of multiple different fabrics coupled to one another to define the barrier.

(42) The protective apparel apparatus **10** may also comprise a face shield **26A**, **26B**. The face shield **26A**, **26B** portion of the protective apparel apparatus **10A**, **10B**, **10C** enables the wearer to see through the barrier provided by the surgical garment **20**. The face shield **26A**, **26B** is generally a sheet-like structure and may have any suitable thickness. The face shield **26** may be mounted and/or attached to an opening or cutout formed in the fabric **24** of the surgical garment **20**. The fabric **24** may be attached around the periphery or edge of the face shield **26A**, **26B** by sewing, snaps, hook and hoop, adhesive, welding, or any combinations thereof. The face shield **26A**, **26B** may be constructed from a transparent material, such as a polycarbonate. The face shield **26A**, **26B** may also be tinted to protect the wearer's eyes from heightened exposure of bright lights. Furthermore, the face shield **26A**, **26B** may be flexible such that the face shield **26** may be curved to accommodate different head sizes. The face shield **26A**, **26B** will be positioned in front of the wearer's face when the surgical garment **20** is disposed over a surgical helmet worn by a medical professional.

(43) Referring to FIG. **2A**, a first configuration of protective apparel apparatus **10A** including a first configuration of a face shield **26A** is illustrated. In this configuration of the protective apparel apparatus **10A**, the face shield **26A** comprises a sheet-like structure that is disposed in an opening in the surgical garment **20**. The opening of the surgical garment **20** couples to the periphery of the shield **26A** such that any holes or apertures in the shield **26A** are covered by the surgical garment **20**, creating a solid sterile barrier between the wearer and the environment. This may include the surgical garment **20** covering an opening in the top of the face shield **26A** that is configured to couple with the surgical helmet **22A**, **22B**. The surgical garment **20** may also cover ferrous or magnetic rivets that are coupled to the face shield **26A** configured to couple with a corresponding fastener on the surgical helmet **22A**, **22B**.

(44) Referring to FIG. **2B**, a second configuration of protective apparel apparatus **10B** including a second configuration of a face shield **26B** is illustrated. The face shield **26B** may comprise one or more openings **28**. For example, the face shield **26B**, as illustrated in FIG. **2B**, may comprise one opening **28**. The opening may be configured to interact with and/or engage the headlamp **12** of the surgical helmet **22** and/or the lens assembly **14**. Alternatively, it is also contemplated that the face

shield **26B** may comprise two openings. For example, the face shield **26B** comprises a first opening **28** for use with the headlamp **12** and/or the lens assembly **14** and a second opening for securing the surgical helmet **22A**, **22B** to the face shield **26B**. In this instance, the protective apparel apparatus **10B** may be disposed over the helmet **22A**, **22B** as illustrated in FIG. **1A**. In one configuration, the surgical helmet **22A**, **22B** may comprise a tab configured to mate with the second opening or aperture in the face shield **26B**. However, the second opening in the face shield **26B** would be covered by the surgical garment, so as not to be exposed.

(45) Referring to FIG. **2C**, an exemplary configuration of a protective apparel apparatus **10A** including a lens assembly **14A** coupled to the face shield **26A**. The lens assembly **14** may be coupled to the protective apparel apparatus **10** such that it is manipulatable by the wearer, on the environment side, to direct and/or manipulate the intensity and/or direction of the light from the headlamp without having to break the sterile barrier. The lens assembly **14** may be manipulated to adjust, tilt, focus, and/or manipulate the headlamp itself in order to direct the light to the surgical workspace. The lens assembly **14** will be described in detail further below.

(46) Referring to FIGS. **3A-3C**, various configurations of the lens assembly **14** for use with the surgical helmet **22** including a headlamp **12** are illustrated. The lens assembly **14** may comprise a housing **48A**, **48B**, **48C**, that is configured to be at least partially disposed on the environment side of the protective apparel apparatus **10A**. The housing **48A**, **48B**, **48C** may be configured to be positioned proximate the face shield **26A**, **26B** and to engage a coupling member **40A**, **40B**, **40C** to attach the lens assembly to the protective apparel apparatus **10A**, **10B**, **10C**. This may be accomplished in a number of different ways depending on the type and or configuration of the protective apparel apparatus **10A**, **10B**, **10C**. For example, the coupling member **40** may be configured to be disposed on and/or coupled to an interior surface on the face shield **26A** on the wearer side of the protective apparel apparatus **10A**. The housing **48** may then be configured to be disposed on and/or coupled to an exterior surface on the face shield **26A** on the wearer side of the protective apparel apparatus **10A**. Alternatively, it is also contemplated that the coupling member **40B** and/or the housing **48B** may be at least partially disposed within the opening **28** in the face shield. For example, a portion of the housing **48B** may be at least partially disposed within the opening **28** of the face shield **26B** such that the control member **18B**, at least partially disposed on the wearer side, may couple to the housing **48B**. In yet another configuration, a portion of the coupling member **40C** may be at least partially disposed within the opening **28** of the face shield **26B** such that the housing **48C**, at least partially disposed on the environment side, may couple to the control member **18C**. Exemplary configurations will be described in greater detail below with regard to FIGS. **3A-3C**.

(47) Referring to FIGS. **3A** and **3B**, the lens assembly **14A**, **14B** may generally comprise a control member **18A**, **18B**. The control member **18A**, **18B** may be disposed on the environment side of the protective apparel apparatus **10A**, **10B** and configured to engage the housing **48A**, **48B** of the lens assembly **14A**, **14B**. The control member **18A**, **18B** may also comprise a lens **58A**, **58B** configured to manipulate the direction and/or intensity of the light passing through the lens assembly **14A**, **14B**. The control member **18A**, **18B** may be manipulatable by the wearer, on the environment side, to direct and/or manipulate the intensity of the light from the headlamp **12** without having to break or compromise the sterile barrier. For example, the control member **18A**, **18B** may be configured to threadably engage the housing **48A**, **48B** such that the control member **18A**, **18B** may be rotated to manipulate the position of the control member **18A**, **18B**, and by extension the lens **58A**, **58B**, relative to the housing **48A**, **48B** and/or the headlamp **12**. Alternatively, the control member **18A**, **18B** may be configured to slidably engage the housing **48A**, **48B** such that the control member **18A**, **18B** may be slid relative to the housing **48A**, **48B** to manipulate the position of the control member **18A**, **18B**, and by extension the lens **58A**, relative to the housing **48A**, **48B** and/or the headlamp **12**. The lens assembly **14A** may be configured such that the manipulating the position of the control member **18A**, **18B**, and by extension the lens **58A**, **58B**, relative to the housing **48A**,

48B and/or the headlamp **12** will adjust the direction and/or intensity of the light passing through the lens assembly **14A**, **14B**. This concept will be explained in greater detail below.

(48) FIG. 3A illustrates a partially exploded view of a first configuration of a lens assembly **14A** configured for coupling to the first configuration of the protective apparel apparatus **10A** of FIG. 2A. The first configuration of the protective apparel apparatus **10A** may comprise the first configuration of the face shield **26A**. As described above, the face shield **26A** does not include any exposed openings or apertures when the face shield **26A** is coupled to the surgical garment **20** in order for the protective apparel apparatus **10A** to form a solid sterile barrier. In such a configuration, a portion of the lens assembly **14A** may be integrally formed with the face shield **26A** and/or coupled to the face shield **26A**. For example, the face shield **26A** may be formed as unitary piece including the features of the housing **48A** formed on the environment side of the face shield **26A**. Wherein the housing **48A** is formed as a portion of the face shield **26A**, the housing **48A** would include a lens **16A** to maintain the sterile barrier created by the protective apparel apparatus **10A**. Alternatively, it is contemplated that, the lens assembly **14** may be attached to the face shield **26A** by an adhesive. In this configuration, a coupling member **40A** may be adhere to the face shield **26A** on the wearer side and housing **48A** of the lens assembly **14** may be adhere to the face shield **26A** on the environment side of the sterile barrier. In yet another configuration, the lens assembly **14** may be attached to the face shield **26A** by magnetic attraction. For example, the face shield **26A** may comprise a placement feature such as a recess or detent on the surface of the face shield **26A** to assist with placement and/or positioning of the lens assembly. Each of the coupling member **40A** and the housing **48A** may have a similar feature configured to engage and/or mate with the placement feature on the respective surfaces of the face shield **26A**. Each of the coupling member **40A** and the housing **48A** may also comprise one of a ferrous material or a magnetic material configured to create a magnetic attraction between the coupling member **40A** and the housing **48A** across the face shield **26A**. For example, the coupling member **40A** may comprise one of a ferrous material and a magnetic material and the housing **48A** may comprise the other of a ferrous material and a magnetic material, such that when both the coupling member **40A** and the housing **48A** are position on their respective sides of the face shield **26A**, the ferrous material and the magnetic material create a magnetic attraction between the coupling member **40A** and the housing **48A** that removably attaches the lens assembly **14A** to the face shield **26A**. Alternatively, it is contemplated that the face shield **26A** may comprise one of one of a ferrous material and a magnetic material imbedding with the face shield **26A**, and the control member **40A** and the housing **48A** may comprise the other of a ferrous material and a magnetic material to create a magnetic attraction between face shield **26A** and the coupling member **40A** and the housing **48A**. The headlamp **12** may then be positioned proximate to and/or coupled to the coupling member **40A** when the protective apparel apparatus **10A** is disposed over the surgical helmet **22**. Additionally, the control member **18A** may be similarly configured to be manipulated in one of the methods described above to manipulate the direction and/or intensity of the light passing through the lens assembly **14A**.

(49) FIG. 3B illustrates a partially exploded view of a second configuration of a lens assembly **14B** configured for coupling to the second configuration of the protective apparel apparatus **10B** of FIG. 2B. As described above, the face shield **26B** may comprise one or more openings or apertures **28B**. The location and shape of the opening **28B**, including any additional openings, may vary. Shape configurations may be any suitable shape such as a circle, oval, square or the like. For instance, the opening **28B** proximate the center portion of the face shield **26B** with a circular shape configuration. Furthermore, the protective apparel apparatus **10B** may include other materials, different from the face shield **26**, disposed within the opening **28B** such that the additional materials of the face shield **26B** spans the opening **28B**. In such a configuration, the lens assembly **14B** may be configured to couple with the face shield **26B** via the opening **28B**. For example, the coupling member **40B** may be configured to couple to, or be partially disposed within, the opening

28B. The headlamp **12** may then be configured to couple with the portion of the coupling member **40B** on the wearer side of the barrier formed by the protective apparel apparatus **10B**. The housing **48B** may then be configured to couple with the portion of the coupling member **40B** disposed within the opening **28B** of the protective apparel apparatus **10B**. In this configuration, the control member **40B** would include a lens or cover **15B** to maintain the sterile barrier created by the protective apparel apparatus **10B**. The control member **18B** may be similarly configured to be coupled to the housing **48B** and manipulated in one of the methods described above to manipulate the direction and/or intensity of the light passing through the lens assembly **14B**.

(50) Referring to FIG. 3C, a partially exploded view of a third configuration of a lens assembly **14C** is illustrated. Similar to the previously described configurations of the lens assembly **14A**, **14B**, the third configuration of the lens assembly **14C** may comprise a housing **48C** configured to couple with a coupling member **40C** to couple the lens assembly **14C** to the face shield **26A**, **26B** of the protective apparel apparatus **10C**. As illustrated in FIG. 3C, the protective apparel apparatus **10C** may be configured such that the coupling member **40C** is integrally formed with the face shield **26C**. In this configuration, the coupling member **40C** would include a lens or cover **15C** to maintain the sterile barrier created by the protective apparel apparatus **10C**. The coupling member **40C** as integrally formed with the face shield **26C** may be configured such that a portion **43** of the coupling member **40C** is disposed on the wearer side of the face shield **26C** to engage and/or couple with the headlamp **12**. The coupling member **40C** as integrally formed with the face shield **26C** may also be configured such that a portion **43C** of the coupling member **40C** is disposed on the environment side of the face shield **26C** to engage and/or couple with the housing **48C** of the lens assembly **14C**.

(51) The lens assembly **14C** may comprise a control member **18C** including a lens **58C** for manipulating the direction and/or intensity of the light passing through the lens assembly **14C**. However, the lens assembly **14C** may also comprise an adjustable portion **17**. The adjustable portion **17** may be disposed between the control member **18C** and the housing **48C** and configured to allow the user to further manipulate the direction of the light passing through the lens assembly **14C**. For example, the adjustable portion **17** may comprise a flexible arm such that allows the user may further manipulate the position and/or orientation of the control member **18C** relative to the housing **48C**, and by extension the direction of the light. The adjustable portion **17** may also comprise an extensible and/or collapsible arm that allows the user to further manipulate the distance between of the control member **18C** and the housing **48** controlling the location that the light may exit the lens assembly **14C**. An example of an adjustable portion **17** may include a semi-rigid sleeve or member with a fiber optic cable **19** at least partial disposed within adjustable portion **17**. The adjustable portion **17** may configured to allow the user to manipulate the position of the control member **18C** to direct and/or angle the light toward the surgical area. The adjustable portion **17** may be configured to keep the control member **18C** and the fiber optic cable in a fixed position relative to the protective apparel apparatus **10C** and/or the surgical helmet **22** until manipulated further by the user.

(52) While various combinations of the lens assembly **14A**, **14B**, **14C**, coupling member **40A**, **40B**, **40C**, and housing **48A**, **48B**, **48C** for use with a face shield **26A**, **26B**, **26C**, they are not intended to be limiting. The combinations described above are merely exemplary configurations of the protective apparel apparatus **10A**, **10B**, **10C**, and additional combinations for coupling the lens assembly **14A**, **14B**, **14C**, coupling member **40A**, **40B**, **40C**, and housing **48A**, **48B**, **48C** with a face shield **26A**, **26B**, **26C** utilizing the components described above are contemplated.

II. Lens Assembly

(53) As described above, in order to maintain the sterile barrier between the wearer and the surrounding environment and to adjust a characteristic of the light during a surgical procedure, the protective apparel system **5** may comprise protective apparel apparatus **10** including a lens assembly **14** at least partially disposed on the environment side of the face shield.

(54) Referring to FIG. 4, a cross-sectional view of a portion of the protective apparel system 5 of FIG. 1A along line A-A is illustrated. The protective apparel system 5 includes the headlamp 12 of the surgical helmet 22 coupled to the protective apparel apparatus 10B including a lens assembly 14B. As described above, the headlamp 12 may include a headlamp body 37. The headlamp body 37 may be formed in any suitable configuration. The headlamp 12 may also comprise a transparent cover 36 and a light source 38 at least partially disposed within the headlamp body 37. The transparent cover 36 may comprise a lens, screen, shield, or other similar cover mechanism configured to allow light to pass through. The transparent cover 36 may be configured to cover a headlamp opening defined by the headlamp body 37 in order to prevent dust, dirt, or moisture from entering into the inside of the headlamp 12, while also allowing light transmitted by the light source 38 to pass through. The transparent cover 36 may comprise any type of material, including, but not limited to, plastic, glass, and/or the like. Additionally, the transparent cover 36 may be a plain screen, without any optically active elements, or a diffuser lens, with optically active elements such as cylindrical lenses, spherical lenses, Fresnel lenses, prisms and the like. In one configuration, the headlamp 12 may comprise any number of screens, lens, and/or reflectors.

(55) The headlamp body 37 may further comprise a mounting feature 42 configured to couple with the coupling member 40. The coupling member 40 may comprise a coupling feature 44 such that the mounting feature 42 of the headlamp 12 and the coupling feature 44 of the coupling member are configured to engage each other when the coupling feature 40 and the headlamp 12 are coupled together. For example, the mounting feature 42 of the headlamp 12 may be configured to matingly engage a complementary coupling feature 44 of the coupling member 40. The mounting feature 44 and the coupling member 40 may be any configuration, including, but not limited to, a snap fit or detent connection, a magnetic connection, threaded connection, and the like. For example, as shown in FIG. 4, a coupling feature 44 may comprise an arm projecting from the coupling member 40 that includes a detent. The detent of the coupling feature 44 may be configured to engage the mounting feature 42 of the headlamp 12 that includes a recess or notch to removably couple the coupling member 40 and headlamp 12.

(56) Alternatively, while not illustrated in the figures, it is also contemplated that the coupling member 40 and/or the lens assembly 14 may be configured to couple to the helmet 22. As illustrated in FIG. 1C, it is contemplated that the helmet 22B may comprise a mounting feature 45. In this configuration, the headlamp 12 may be mounted to the mounting feature 45 of the helmet 22B as described above. The coupling feature 44 of coupling member 40 may then be configured to be coupled directly to the headlamp 12, to the mounting feature 45 of the helmet 22B, or some combination thereof, such that the headlamp 12 is aligned with the lens assembly 14 when the coupling member 40 and/or the lens assembly 14 is mounted to the helmet 22B via the mounting feature 45. For example, the coupling feature 44 of the coupling member 40 may be configured to mate with the mounting feature 45 of the helmet 22B such that the coupling member 40 is coupled to the helmet 22. In this configuration, a headlamp 12 may also couple to the mounting feature 45 of the helmet 22B such that the coupling member 40 and the headlamp 12 are aligned. In such a configuration, the coupling member 40 may abut the inner surface of the face shield 26 when the protective apparel apparatus 10 is disposed over the helmet 22B. It is also contemplated that the headlamp 12 may be removably coupled to the mounting feature 45 of the helmet 22B, and the coupling member 40 of the lens assembly 14 may be coupled to the headlamp 12 in a similar manner as described above with regard to headlamp 12 and lens assembly 14 of FIG. 4.

(57) In the configurations contemplated above, the coupling member 40 may be disposable and/or reusable. For example, the coupling member 40 may be removably coupled to allow for disposal of the protective apparel apparatus 10 and reuse of the surgical helmet 22 following a procedure or exam. Alternatively, it is also contemplated that the coupling member 40 may be permanently coupled to the protective apparel apparatus 10 and is disposed of following a procedure or exam.

(58) The lens assembly 14 may also comprise a housing 48 and a control member 18 disposed on

the environment side of the protective apparel apparatus **10**. The housing **48** may be integrated with or permanently affixed to the face shield **26** in any suitable method. For example, the housing **48** may be integrated as a single unit with the face shield **26** of the surgical garment **20**. In this configuration, the housing **48** may be coupled to the environment side of the face shield **26** using an adhesive such that the housing **48** will be aligned with the headlamp **12**/coupling member **40** on the wearer side of the surgical garment **20** when the protective apparel apparatus **10** is disposed over the helmet **22**. Suitable methods of permanently coupling the housing **48** of the lens assembly **14** to the environment side of the shield **26** may include, but is not limited to, welding, adhesion, or the like. In this configuration, other components of the lens assembly **14** may be removable. For example, the control member **18** may be removed and reused. Alternatively, it is also contemplated that the housing **48** may be removably coupled to the protective apparel apparatus **10**. For example, the housing **48** may be coupled to the protective apparel apparatus **10** via an intermediary connecting material such as a hook and loop material, straps, or by snaps and the like. In this configuration, a hook portion may be disposed on the housing **48** and a loop portion may be disposed on the face shield **26**, or vice versa. It is also contemplated that in the configuration of the protective apparel apparatus **10B** wherein the face shield **26B** including the opening **28**, the housing **48B** may comprise a retention feature configured to create a friction fit or other similar connection with the opening **28B** in the face shield **26B**. In yet another configuration, the housing **48B** may comprise a retention feature configured to extend through the opening **28B** in the face shield **26B** and coupled to the coupling member **40B**.

(59) In configurations wherein the face shield **26B** comprises an opening **28**, the housing **48B** and/or the coupling member **40B** may comprise a lens or cover to provide a solid barrier with the face shield **26B**. For example, as illustrated in FIG. 4, the housing **48** may be permanently coupled to the face shield **26** and comprise a lens **16** so that there are not breaks or openings in the barrier provided by the protective apparel apparatus **10**. In this configuration, the coupling member **40** may be removably coupled to the housing **48** and/or the face shield **26**. Alternatively, while illustrated in FIG. 4, it is also contemplated that coupling member **40** may comprise a lens/cover and be permanently coupled to the face shield **26** so that there are not breaks or opening in the barrier provided by the protective apparel apparatus **10**. In this configuration, the housing **48** may be removably coupled to the coupling member **40** and/or the face shield **26B**.

(60) The lens assembly **14** may further comprise one more lenses **16**, **58**. As illustrated in FIG. 4, the lens assembly **14** may comprise a first lens **16** and a second lens **58**. The first lens **16** may be coupled to the housing **48**, and the second lens **58** may be coupled to the control member **18** of the lens assembly **14**. The first lens **16** and the second lens **58** may be configured to fit snugly in the housing **48** and the control member **18**, respectively. As described above, the control member **18** may be configured to be manipulatable by the user. The control member **18** and the housing **48** may be coupled together such that manipulation of the control member **18** moves the control member **18** and, by extension, the second lens **58**, relative to the housing **48** and, by extension, the first lens **16**. The control member **18** may be configured such that the wearer may easily manipulate the control member **18** to adjust a position of the second lens **58** relative to the first lens **16** by rotating, sliding, or similarly manipulating the control member **18**.

(61) Depending on the type of manipulation of the control member **18** and/or the coupling between the housing **48** and the control member **18**, a characteristic of the lens assembly **14** is adjusted. In one example, manipulation of the control member **18** may vary the distance or position of the first lens **16** and/or the second lens **58**. Varying a distance from the first lens **16** and the second lens **58** varies the light from the headlamp **12** to a desired focal point **74**. Alternatively, varying a distance from the first lens **16** and the second lens **58** may vary the light path from the headlamp **12**. The desired focal point **74** may include, but not limited to, the workspace or an area of interest. Alternatively, varying a distance or position of the lens **16**, **58** may vary the light path from the headlamp **12**. In another example, manipulation of the control member **18** may vary an angle of the

light emitted from the headlamp. In this example, the housing **48** may comprise reflectors, fiber optic cables, and the like. In an exemplary configuration, as illustrated in FIG. 3C, the lens assembly **14** may be extendable or movable such that manipulation of the control member **18** by the wearer varies the light angle. In this configuration, the housing **48** may comprise fiber optic cables, reflectors or the like to aid in varying light angle. For instance, in combination with the use of any number of reflective surfaces or fiber optic cables, the light from the headlamp **12** may be redirected or articulated such that the surgical field, workspace, and/or patient is illuminated. It will be appreciated that the other characteristics of the lens assembly **14** may include, but not limited to, adjusting the total internal reflection of the lens assembly **14**.

(62) While the sectional view of FIG. 4 illustrates an exemplary lens assembly **14** including a first lens **16** and a second lens **58**, it is contemplated that the lens assembly **14** may comprise any number of lenses. For example, while not illustrated in the figures, it is contemplated that the lens assembly **14** may comprise only one (first) lens **16**. Furthermore, it is contemplated the system **5** may include any number of additional lenses. For example, the lens assembly **14** may also comprise a screen or cover similar to the transparent cover **36** of the headlamp **12** described above. It is also contemplated that the headlamp **12** may include additional lenses, such as the transparent cover **36**, wherein the combination of the headlamp **12**, lens assembly **14** and the transparent shield **26** comprise a system/assembly including a plurality of lenses working together to modify the direction and/or intensity of the light produced by the headlamp **12**. In some configurations, the “first” lens and the “second” lens may be interchangeable. For example, the headlamp **12** may comprise the “first” lens and the lens assembly **14** may comprise the “second” lens and vice versa.

(63) Referring to FIG. 5A-6B, two exemplary configurations of the lens assembly **14A**, **14B** for use with a headlamp **12** are illustrated. As described above, the lens assembly **14A**, **14B** may comprise a coupling member **40A**, **40B**. The coupling member **40A**, **40B** may comprise any suitable shape, such as circular, oval, or polygonal shapes. The coupling member **40A**, **40B** may include a recess having various shapes and/or sizes in which light generated from the headlamp **12** may pass through.

(64) Referring to FIGS. 5A and 5B, a first configuration of the lens assembly **14A** and a first configuration of the coupling member **40A** wherein the coupling member **40B** may have an inner portion **41A** and an outer portion **43A** with the inner portion **41A** including a mounting feature **44A** for coupling to the headlamp **12**. The coupling member **40A** may be configured to be couple with the first configuration of the protective apparel apparatus **10A** including the face shield **26A**, as illustrated in FIGS. 2A and 3A. In this configuration the face shield **26A** does not include any exposed openings. The outer portion **43A** of the coupling member **40A** may abut the wear side of the face shield **26A**. As described above, the coupling member **40A** may comprise a retention feature **39A** for coupling the coupling member **40A** to the face shield **26A**. The retention feature **39A** may comprise a magnet, hook and loop, adhesive, or similar fastener. The face shield and/or housing **48A** may then comprise a corresponding retention feature **48A**. For example, the coupling member **40A** may comprise a retention feature **39** including a ferrous material and the housing **48A** may comprise a retention feature **49** including a magnetic material, such that when both the coupling member **40A** and the housing **48A** are position on their respective sides of the face shield **26A**, the ferrous material and the magnetic material create a magnetic attraction between the coupling member **40A** and the housing **48A** that removably attaches the lens assembly to the face shield **26A**. Alternatively, it is contemplated that the coupling member **40A** may comprise a retention feature **39A** including a magnetic material and the face shield may comprise a ferrous material imbedding with the face shield **26A** such that the ferrous material and the magnetic material create a magnetic attraction between the coupling member **40A** and the face shield **26A**. In this configuration, the housing **48A** may similarly comprise a retention feature **49** including a magnetic material such that the ferrous material imbedded in the face shield **26A** and the magnetic material of the housing **48A** create a magnetic attraction between the housing **48A** and the face

shield 26A.

(65) Referring to FIGS. 6A and 6B, a second configuration of the lens assembly 14B and a second configuration of the coupling member 40B wherein the coupling member 40B may have an inner portion 41B and an outer portion 43B with the inner portion 41B including a mounting feature 44B for coupling to the headlamp 12. The coupling member 40B may be configured to be couple with the second configuration of the protective apparel apparatus 10B including the face shield 26B with an opening 28B, as illustrated in FIGS. 2B and 3B. The outer portion 43B of the coupling member 40B may abut the wear side of the face shield 26B and the aperture defined by the coupling member 40B may generally encircle the opening 28B. Alternatively, the coupling member 40B need not encircle the opening 28B. For example, the coupling member 40B may comprise one or more individual tabs coupled to the wearer side of the face shield 26B proximate to the opening 28B. The housing 48B may be coupled to the face shield 26B and comprise a lens configured to cover the opening. The plurality of tabs may be configured to couple to the headlamp 12 in order to align the headlamp with the housing 48 and/or the lens assembly 14B. The coupling member 40B may have other configurations other than what is shown in the figures and described herein.

(66) As mentioned above, the coupling member 40 may couple to the headlamp 12, the helmet 22, and/or the face shield 26. It is also contemplated that the coupling member 40 is configured to couple to the lens assembly 14. In each of the configurations described above, it is contemplated that the coupling member 40 and/or the lens assembly 14 may be disposable and/or reusable. For example, the coupling member 40 may be removably coupled to allow for disposal of the surgical garment 20 and face shield 26 portions of the protective apparel apparatus 10 and reuse of the coupling member 40 following a sterilization procedure/process. Alternatively, the coupling member 40 may be permanently coupled to the face shield 26 such the coupling member 40 is disposed of with the protective apparel apparatus 10 following a procedure or exam. It is also contemplated that portions of the lens assembly 14 may be reusable while other portions are disposable. For example, the housing 48 and/or control member 18, which may be removable in certain configurations of the protective apparel apparatus 10, may be configured to be reusable. Alternatively, the housing 48 portion of the lens assembly 14 may be permanently coupled to the protective apparel apparatus 10 and may be disposed of with the surgical garment 20 and face shield 26 portions of the protective apparel apparatus 10.

(67) In configurations wherein a solid barrier is provided, the lens assembly 14 may be reusable with another surgical garment 20. Referring to FIG. 2C, the wearer may use the lens assembly 14 with one surgical garment 20, remove the lens assembly 14, and then use the lens assembly 14 with another surgical garment 20. It will be appreciated that any or all components of the lens assembly 14 may be reusable. Additionally, the lens assembly 14 may be sterilized between each usage and/or removal of the lens assembly 14 in order to maintain the sterile barrier.

(68) In some configurations wherein the face shield 26B comprises an opening 28B, as shown in FIG. 2B, the coupling member 40 may comprise a cover similar to the transparent cover 36 of the headlamp 12 as mentioned above or a lens similar to the first lens 16. The coupling member 40 in combination with the cover or lens may form the sterile barrier in conjunction with the face shield 26 and the rest of the surgical garment 20. Further, in these configurations, the coupling member 40 is attached to the face shield 26. In other configurations, the housing 48 in combination with the first lens 16 may form the sterile barrier with the face shield 26. In these configurations, components such as the control member 18 of the lens assembly 14 may be reusable and/or removable. Referring to FIG. 3B, the housing 48B may be constructed and attached to the surgical garment 20B about the opening 28B such that bacteria cannot enter or exit through the opening 28B once the housing 48 abuts and/or is coupled to the face shield 26B. In these configurations, the lens assembly 14 may be reusable and/or removable without breaching the sterile barrier.

(69) In yet other configurations wherein the face shield 26 may not comprise any openings, as shown in FIG. 2A and 3A, the face shield 26 provides a solid barrier between the environment side

and the wearer side. In these configurations, any or all components of the lens assembly **14** may be reusable, removable, and/or disposable as mentioned above. Removing any of the aforementioned components will not break the sterile barrier between the environment side and the wearer side as a solid barrier is provided by the face shield **26**, the coupling member **40**, and/or the headlamp **12**.

(70) The sterile barrier is arranged between the wearer side and the environment side of the surgical garment **20**. Any microbes or other contamination (e.g. dust, particulate matter, other biological matter, etc.) that is introduced during a surgical procedure will remain on the environment side of the surgical garment **20** and/or remain on the wearer side of the surgical garment **20**.

(71) In certain configurations, a significant advantage of the lens assembly **14** is that the sterile barrier is designed as a continuous, solid barrier that is maintained throughout any manipulation of the control member **18** of the lens assembly **14**. Such a solid barrier can be provided because, in one example, the housing **48** in combination with the lens **16** of the lens assembly **14** is adapted to form a part of the sterile barrier such that the sterile barrier may extend unbroken between the housing **48** and the face shield **26**. Thus, any manipulation of the lens assembly **14** during a surgical procedure does not compromise the sterile barrier between the wearer and the environment.

(72) As mentioned above, the coupling member **40** may be coupled to the lens assembly **14** by an attachment feature **46**. More specifically, the coupling member **40** may be coupled to the housing **48** of the lens assembly **14** with the attachment feature **46**. The housing **48** may comprise a (mating) attachment feature **50** such that the attachment feature **46** and the mating attachment feature **50** are attached to each other. For example, as shown in FIG. **4**, the attachment feature **46** may comprise a ferrous material and the mating attachment feature **50** of the housing **48** is a magnet, or vice versa. When the attachment feature **46** contains material that is capable of being attracted to a magnetic material, the mating attachment feature **50** of the housing **48** can, in addition or alternatively, include magnetic components. In other words, the attachment feature **50** may comprise a permanent magnetic material and the complementary element, in this example is attachment feature **46** of the coupling member **40**, comprises a ferrous material or vice versa. In this manner, the housing **48** of the lens assembly **14** can be secured to the coupling member **40**, at least in part, magnetically.

(73) The mating attachment feature **50** may be coupled to any component of the lens assembly **14** such that mating attachment feature **50** is attached to the attachment feature **46** of the coupling member **40**. Further, the attachment feature **50** may be attached to the coupling member **40** in any suitable manner such that the sterile barrier between the wearer side and the environment side is maintained.

(74) Another example, the attachment feature(s) may comprise a press or shrink fit whereby there is an interference fit between the coupling member **40** and the housing **48** in which one of the two elements is forced under pressure into the other element through the surgical garment in a way sufficient to maintain a sterile barrier. It will be appreciated that the attachment feature(s) **46**, **50** may be any type of attachment, including, but not limited to, threading, adhering, mechanical fastening, and/or pressure fitting. Alternatively, both the coupling member **40** and the housing **48** may be permanently attached to the face shield **26**.

(75) In some configurations, the mounting feature **44** of the coupling member **40** and the attachment feature **46** may be a single unit or component such that the mounting feature **44** and the attachment feature **46** are the same feature configured to mount both the headlamp **12** and the lens assembly **14** simultaneously. For example, the coupling member **40** is a magnet. The magnet may be positioned proximate to the opening **28**. The mounting feature **44** and the attachment feature **46** mounts to the headlamp **12** and the lens assembly **14**, respectively, through magnetic force. In other words, the mounting feature **44** and the attachment feature **46** is the same feature (magnetic force) of the coupling member **40**. Alternatively, the mounting feature **44** and the attachment feature **46** may be different and/or separate features of the coupling member **40**.

(76) Furthermore, the housing **48** is configured to couple to the control member **18** as shown

throughout the Figures. The housing **48** may couple to the control member **18** in any suitable method. The control member **18** may be manipulated by the wearer to move relative to the housing **48**. Various configurations of the control member **18** and techniques of manipulation are contemplated in detail below.

(77) In one configuration, as shown in FIG. 6B, the housing **48** may comprise a plurality of external threads **52** at one axial end thereof for threadably engaging internal threads **54** in a cavity of the control member **18**. The external and internal threads **52**, **54** may be right-handed threads or left-handed threads. In this configuration, the wearer may manipulate the control member **18** by rotating the control member **18** relative to the housing **48** such that the external threads **52** engages the internal threads **54** of the control member **18** whereby moving the control member **18** relative to the housing **48**. Depending on the direction of the rotation, the control member **18** may be manipulated to move towards or away from the housing **48**.

(78) In another configuration, the control member **18** may include a motion or gesture detector. For example, to manipulate the control member **18**, the wearer may perform a motion such as a finger swipe on the control member **18**. In this configuration, the control member **18** may include any number of sensors or detectors.

(79) In another configuration, the control member **18** may be slidably engaged relative to the housing **48** of the lens assembly **14**. In other words, to manipulate the control member **18**, the wearer may slide an element of the control member **18** or the entirety of the control member **18** relative to the housing. For example, the control member **18** may be slidably engaging the housing **48** such that the wearer may slide the control member **18** forwards, backwards, upwards, or downwards relative to the housing **48** to alter one or more characteristics of the lens assembly **14**.

(80) In another configuration, the control member **18** may take the form of a handle, lever, button or the like. To manipulate the control member **18**, the wearer may apply force or pressure thereto. For example, the control member **18** may include directional buttons wherein if the wearer presses a directional button, a component of the lens assembly **14** may move to the corresponding direction to alter a characteristic of the lens assembly **14**.

(81) In another configuration, the control member **18** may be remotely manipulatable via a controller. Additionally, an input device may be in communication with the control member **18** and the controller such that the input device may be utilized to manipulate the control member **18**. Input devices may include, but not limited to, a display screen with touch screen capabilities, a joystick, a keyboard, a microphone, and a scanner.

(82) In another configuration, the housing **48** may be formed such that the housing **48** and the control member **18** are not readily separable from one another once assembled. For example, the housing **48** may comprise retainer posts adapted to snap fit within the cavity of the control member **18**.

(83) In yet another configuration, the housing **48** may include the control member **18** such that the housing **48** and the control member **18** are integrated into a single component or unit. In this example, the housing **48** may be integrated with the control member **18** during manufacture of the lens assembly **14**. Alternatively, the housing **48** may be separately manufactured from the control member **18** and wherein once the housing **48** engages the control member **18**, the components **18**, **48** are inseparable.

(84) Additionally, the control member **18** may comprise a gripping feature **56** designed to enable the wearer to easily manipulate the control member **18**. For example, the gripping feature **56** may be grooves formed on the control member **18**. In operation, the wearer may grip the control member **18** along the grooves **56** and manually couple the control member **18** to the housing **48**. The wearer may rotate the control member **18** towards to the housing **48** such that, in one configuration, the internal threads **54** of the control member **18** threadably engage the external threads **52** of the housing **48**.

(85) Referring to FIGS. 7A and 7B, the first lens **16** and the second lens **58** of the lens assembly **14**

may each independently comprise any number of surfaces of any varying forms, including, but not limited to, convex, concave, and concave-convex (shaped) surfaces. As shown throughout the Figures, the first lens **16** may comprise two surfaces **60**, **62** and the second lens **58** may comprise two surfaces **64**, **66**. In one configuration, the surface **60** of the first lens **16** may be a convex surface and the surface **62** may be a concave surface or vice versa. In another configuration, the surface **60** of the first lens **16** may be a convex surface and the surface **62** may be a flat surface or vice versa. As illustrated in FIGS. 7A and 7B for exemplary purposes, in another configuration, the first lens **16** is a dual concave lens with surfaces **60**, **62** curving inwards and the second lens **58** is a dual convex lens with surfaces **64**, **66** curving outwards. The surfaces **60**, **62**, **64**, **66** of the first and second lenses **16**, **58** may be any combination of surfaces.

(86) The first lens **16** and/or the second lens **58** may be made in form of a reflector (e.g. mirror) which reflects the light by common reflection, or as a translucent body made of glass or plastic material (optical lens) where the focusing of light is achieved by optical refraction. In one configuration, the first and second lenses **16**, **58** comprise a lens material **68** formed of a translucent body of glass or plastic material. The lens material **68** may be stiff or flexible. Additionally, the material **68** may be made of acrylate or polystyrene, or comparable low-cost but durable transparent material having suitable refractive properties.

(87) A film **70** and/or a coating **72** may be disposed on any of the surfaces **60**, **62**, **64**, **66** or any other surface of the first and second lenses **16**, **58**. Alternatively, the film **70** may be disposed between the surfaces **60**, **62**, **64**, **66** of the first and/or second lenses **16**, **58**. As shown in FIGS. 7A and 7B, according to one example, both the surfaces **60**, **62**, **64**, **66** of the first and second lenses **16**, **58** are layered with the film **70** and the coating **72**. The surfaces **60**, **62**, **64**, **66** may include any number of layers of film **70** and/or coating **72** in which the layers of film **70** and/or coating **72** may vary in thickness. It will be appreciated that the film **70** and/or the coating **72** may be on any surfaces **60**, **62**, **64**, **66** of the first lens **16** and/or second lens **58**. For example, the film **70** and coating **72** may be on only one of the surfaces **60**, **62** of the first lens **16**. It will further be appreciated that the layering order of the film **70** and the coating **72** may vary depending on the optimal layering order desired.

(88) The film **70** may be of plastic material or glass and can be a plain sheet, without any optically active elements, or a diffuser lens, with optically active elements such as a cylindrical lenses or prisms. Additionally, the film **70** and/or the coating **72** may alter the way in which the lenses reflects and transmits light from the headlamp **12**. Further, the film **70** may be any type of film, including, but not limited to, polarized film and metalized film.

(89) The coating **72** may be any type of coating, including, but not limited to, tint coating, gradient coating, glass coating, plastic coating, anti-reflective coating, scratch-resistant coating, ultra-violet coating, hydrophobic coating, color coating, and mirror coating. The coating may be coated by any conventional method, such as roll coating, gravure coating, bar coating, or extrusion coating. In some configurations, the film **70** and the coating **72** may be understood as a single layer wherein the coating **72** is a coating composition for forming the film **70**.

(90) Referring to FIGS. 8A-9C, an exemplary configuration of the lens assembly **14** is illustrated with the control member **18** at various positions relative to the housing **48** to illustrate the operation of the lens assembly **14**. First, the housing **48** and the control member **18** are positioned at a first distance, D , from each other in FIG. 8A. Then, the wearer may utilize the grooves **56** of the control member **18** to manipulate position of the control member **18** relative to the housing **48**. For example, when the control member **18** is coupled to the housing via a threaded connection, the user may twist/rotate the control member **18** which adjusts the distance between the control member **18** and the housing **48**. FIG. 8B illustrates the control member **18** positioned at a second distance D' relative the housing **48**. FIG. 8C illustrates the control member **18** positioned at a second distance D'' relative the housing **48**. The control member **18** may be manipulated to position of the control member **18** at any number of distances from the housing **48** based on the type of connection

between the control member **18** and the housing **48**. In turn, the manipulation of the control member **18** varies the distance between the first lens **16** and the second lens **58**, as shown in FIGS. **9A-9C**, which redirects or adjusts the light from the headlamp **12**.

(91) Possible paths of the light from the headlamp **12** through the lens assembly **14** caused by the manipulation of the control member **18** by the wearer are depicted in FIGS. **9A-9C**. The distance variation between the housing **48** and the control member **18** is illustrated as a distance D , a distance D' , and a distance D'' and may translate to the distance between the first lens **16** and the second lens **58** illustrated as a distance d , a distance d' and a distance d'' , respectively. In other words, the position of the second lens **58** is adjusted relative to the first lens **16**. FIG. **8C** illustrates further manipulation of the control member **18** translating to a distance variation of first lens **16** and the second **58** to a distance d'' which is shown to redirect light towards a desired focal point **74** or to provide better visibility of the workspace. As illustrated in FIG. **9A-9C**, light is emitted from the light source **38**. The emitted light may pass through a lens **36** of the headlamp **12**. The light may then pass through a lens **16** disposed within the coupling member **40** and/or the housing **48**. The lens **16** may be configured to collect and/or direct the emitted light to through the lens assembly **14** toward a second lens, such as the lens **58** of the control member **18** of the lens assembly **14**. The lens **58** may then direct the light toward in a direction based on the position of the control member **58** relative to the lens **16** of the coupling member **40** and/or the housing **48**. The shape of the lenses **16**, **58**, and the distance between the lenses may dictate the direction and/or intensity of the light at the focal point **74**.

(92) Referring to FIG. **10**, an exemplary configuration of path of the light from the headlamp **12** through the lens assembly **14C** of FIG. **3C** is illustrated. As illustrated in FIG. **10**, light is emitted from the light source **38**. The emitted light may then be collected by and/or pass through a lens **16** disposed within the coupling member **40C** and/or the housing **48C**. The lens **16** may be configured to direct the collected light to a fiber optic cable **19** configured to transmit the light along the length and/or the path of the fiber optic cable. The fiber optic cable **19** may comprise be configured to terminate at a lens, such as the lens **58C** of the control member **18C** of the lens assembly **14C**. The lens **58C** may then direct the light toward in a direction determine based on the position of the control member **18C**.

(93) The lens assembly **14** increases the efficiency, visibility, and sterility of a surgical procedure by enabling the wearer to redirect the light from the headlamp **12** during surgery without breaching the sterile barrier. Further, the wearer would not need to breach the sterile barrier to adjust the light by tilting the headlamp; rather the wearer would redirect the light from the headlamp **12** as described herein. Moreover, the lens assembly **14** is positioned such that the wearer's view of the surgical field, workspace, and the patient is unobstructed.

(94) The above are directed to specific configurations of the system **5**. It should be understood that the individual features of the different configurations of the system **5** may be combined to construct alternative configurations of the system **5**.

(95) It will be further appreciated that the terms “include,” “includes,” and “including” have the same meaning as the terms “comprise,” “comprises,” and “comprising.”

(96) Several configurations have been discussed in the foregoing description. However, the configurations discussed herein are not intended to be exhaustive or limit the invention to any particular form. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the invention may be practiced otherwise than as specifically described.

Claims

1. A protective apparel system comprising: a surgical helmet to be worn over a head of a wearer, the surgical helmet comprising a peripheral device; a surgical garment configured to be at least

- partially disposed over the surgical helmet defining a microbial barrier around at least the surgical helmet and the head of the wearer, said surgical garment comprising: a surgical fabric; a face shield coupled to the surgical fabric, the combination of the surgical fabric and the face shield configured to define an environment side and a wearer side; and a coupling member coupled to the wearer side of the face shield, the coupling member comprising a coupling feature configured for removable coupling of the peripheral device to the wearer side of the face shield to orient the peripheral device relative to the face shield when the surgical garment is disposed over the surgical helmet.
2. The protective apparel system of claim 1, wherein the coupling member is magnetically coupled to the face shield.
 3. The protective apparel system of claim 1, further comprising a lens assembly; wherein the coupling member further comprises an attachment member; and wherein the attachment member of the coupling member is configured to removably couple the lens assembly to the face shield.
 4. The protective apparel system of claim 3, further comprising a control member, the control member is operatively coupled to the lens assembly such that the control member is configured to be manipulated by the wearer to adjust a characteristic of the lens assembly.
 5. The protective apparel system of claim 3, wherein the face shield further comprises an opening; wherein the lens assembly further comprises a housing, the housing at least partially disposed on the environment side of the face shield; and wherein the housing is coupled to the coupling member disposed on the wearer side of the face shield through the opening such that the face shield is at least partially disposed between the coupling member and the housing.
 6. The protective apparel system of claim 1, wherein the coupling member is at least partially formed as a unitary component of the face shield.
 7. The protective apparel system of claim 1, wherein the face shield further comprises an opening; and wherein the coupling member is at least partially disposed within the opening.
 8. A surgical garment for use with a surgical helmet having a peripheral device, the surgical garment comprising: a surgical fabric configured to provide a microbial barrier between a medical environment and a wearer; a face shield coupled to the surgical fabric, the combination of the surgical fabric and the face shield defining a microbial barrier configured to be around at least the surgical helmet and a head of the wearer, the microbial barrier having an environment side and a wearer side; and a coupling member coupled to the face shield on the wearer side, the coupling member comprising a magnetic coupling feature configured for removable coupling of the peripheral device to the wearer side of the face shield to orient the peripheral device relative to the surgical face shield when the surgical garment is disposed over the surgical helmet.
 9. The surgical garment of claim 8, wherein the coupling member is at least partially formed as a unitary component of the face shield.
 10. The surgical garment of claim 8, wherein the face shield further comprises an opening; and wherein the coupling member is at least partially disposed within the opening.
 11. The surgical garment of claim 8, wherein the coupling member is coupled to the face shield by an adhesive.
 12. The surgical garment of claim 8, wherein the coupling member is magnetically coupled to the face shield.
 13. The surgical garment of claim 8, wherein the coupling member is configured to couple directly to the peripheral device.
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